Astrophysics II – High Energy Astrophysics

Fall Semester 2024

Course and Contact Information

Department and Course:	Physics, PHYS 6730
Semester:	Fall 2024
Time:	Monday and Wednesday 3:45 PM - 5:00 PM
Location:	Corcoran 309
Course Web Site:	http://blackboard.gwu.edu

Zoom feed link: This course is in-person. The link will only be available for emergency use and will be shared as needed.

Instructor: Prof. Oleg Kargaltsev
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Phone: 202-994-7225 (office)
E-Mail: kargaltsev@gwu.edu
Office Hours: Wednesday, 11 am -noon

PHYS 6730— Astrophysics II

Week	Date	Homework Due or Exam	Topics
1	Monday August 26		Course structure. High-energy universe: tour of high-energy phenomena.
1	Wednesday August 28		Review of radiation processes (covered in Astrophysics I).
2	Monday Sep 2		LABOR DAY
2	Wednesday Sep 4		Basic gas/fluid dynamics (I).
3	Monday Sep 9		Basic gas/fluid dynamics (II).
3	Wednesday Sep 11		Plasma physics.
4	Monday Sep 16		Particle motion. Basic magnetohydrodynamics.
4	Wednesday Sep 18	HW#1	Reconnection. Dynamo.
5	Monday Sep 23		Shock waves.
5	Wednesday Sep 25		Particle acceleration
6	Monday Sep 30		Accretion I
6	Wednesday Oct 2		Accretion II
7	Monday Oct 7		Lives of stars and related physical processes.
7	Wednesday Oct 9		White Dwarfs and Isolated Neutron Stars
8	Monday Oct 14	HW#2	Pulsars. Pair production. Pulsar winds.
8	Wednesday Oct 16		Diversity of Neutron Stars (isolated and in binaries).
9	Monday Oct 21		Basics of Black Holes.
9	Wednesday Oct 23		BHs in X-ray binaries.
10	Monday Oct 28	HW#3	Supernovae and Supernova Remnants.

10	Wednesday Oct 30	MIDTERM	
11	Monday Nov 4		Active Galactic Nuclei (AGN).
11	Wednesday Nov 6		Cosmic Rays.
12	Monday Nov 11		TDEs. Gamma-ray bursts.
12	Wednesday Nov 13	Proposal outline is due	Gamma-ray bursts.
13	Monday Nov 18		Gravitational waves and their sources.
13	Wednesday Nov 20		Extracting physics from data: X-ray astronomy example
14	Monday Nov 25		Thanksgiving break
14	Wednesday Nov 27		Thanksgiving break
15	Monday Dec 2	Proposal Progress Report. HW#4 due.	Feasibility section in your proposals. Discussion of your observing proposals.
15	Wednesday Dec 4		NO CLASS (CXO Conference)
16	**Monday Dec 9**	FINAL FROJECT	Proposal presentations and discussions.

Detailed Course Description

LEARNING OBJECTIVES:

In this course, you will:

1. Acquire knowledge about high-energy cosmic sources;
2. Develop analytical, scientific, and critical thinking skills;
3. Develop the ability to identify the appropriate physics laws/principles and mathematical methods needed to tackle a specific astrophysics problem;
4. Make connections between mathematics, physics and astronomy.

COURSE DESCRIPTION: In this survey course of modern high-energy astrophysics, you will learn about various high-energy processes, the 4th state of matter (plasma), extreme states of matter and space (neutron stars and black holes), life and death of massive stars, modern X-ray and gamma-ray observatories, recent developments in gravitational wave and neutrino astronomy, and basics of statistical data analysis.

PREREQUISITE: **Electrodynamics and Classical Field Theory (PHYS 6210).** Astrophysics I (PHYS 6630) is not a required prerequisite, but it will be useful if you already took it.

LECTURES: Lectures are primarily meant to explain difficult concepts, to expand on the reading material, and to introduce topics not covered in the textbooks such as examples from real observations and data or recently published results. If you do not understand something, you are very much encouraged to ask questions during the lectures. If something is not clear, please raise your hand and tell me right away. All previous lectures, all reading assignments, and all homework assignments will be posted in the Blackboad course page. Please check this page frequently. If the page is updated, announcements will be made through the Blackboard.

YOUR RESPONSIBILITIES:

1. Reading the textbook material assigned for every class.
2. Reading any additional materials distributed in class or via Blackboard.
3. Actively participating in discussions during class.
4. Reproducing any calculations and derivations shown in class alone by yourself (with textbook and lecture notes closed).
5. Doing the homework by yourself and turning it in time.
6. Develop an observing proposal (in lieu of final exam).

READING ASSIGNMENTS AND USEFUL CLASSIC BOOKS:

HEA is a very rapidly developing area of astrophysics. In this course we will use most recent research results for various topics we cover and also review some fundamentals. These are several literature free sources you will be **required** to read for this course. Specific reading assignment for each source will be given.

Free notes and books on HEA:

https://arxiv.org/pdf/1703.05575.pdf (by Dr. Luigi Foschini, 2021)
https://dlakaplan.github.io/phys718/lectures.pdf (by David Kaplan, 2019)
https://www.astro.princeton.edu/~jeremy/heap.pdf (by Dr. Jeremy Goodman, 2013)
https://arxiv.org/pdf/1202.5949.pdf (by Dr. Gabriele Ghisellini, 2012)

You will be also **required** to read several recent review articles about particular types of objects. They will be given to you by the instructor.

Taken together, these provide a very good and modern introduction to high-energy astrophysics and X-rya and gamma-ray astronomy.

There are other ***optional*** but good books on high-energy astrophysics that can be useful for various parts of this course and also for your future research. Here are some titles:

* Frontiers of X-ray Astronomy
A.C. Fabian, K.A. Pounds, and R.D. Blandford
Cambridge University Press, 2004
ISBN: 0-521-53487-9
Good reviews on selected topics in X-ray astronomy.

* Exploring the X-ray Universe: Second Edition (**few of these are available for free!**)
F.D. Seward and P.A. Charles
Cambridge University Press, 2010
ISBN: 978-0-521-88483-9

* Accretion Power in Astrophysics: Third Edition
J. Frank, A. King, and D. Raine
Cambridge University Press, 2002
ISBN: 0-521-620538
The book has excellent coverage of accretion physics.

* High-Energy Astrophysics: Third Edition
M.S. Longair
Cambridge University Press, 2011
ISBN: 978-0-521-75618-1
Extensive coverage of high-energy astrophysics including phenomenological astronomy, physical processes, Galactic and extragalactic sources.

* Plasma Astrophysics: Saas-Fee Advanced Course 24. Lecture Notes 1994.
J.G. Kirk, D.B. Melrose, E.R. Priest (Author)
Springer-Verlag Berlin Heidelberg 2010
ISBN: 978-364-208202-3
An excellent, concise introduction to plasma astrophysics and magnetohydrodynamics

* Radiative Processes in Astrophysics
G.B. Rybicki and A.P. Lightman
Wiley Interscience, 1979
ISBN: 0-471-82759-2
This excellent book is frequently used as a graduate level textbook.

* High Energy Radiation from Black Holes: Gamma Rays, Cosmic Rays, and Neutrinos
Charles D. Dermer & Govind Menon
Princeton University Press, 2009
ISBN: 9780691144085
Great modern graduate level textbook covering broad range of topics in theoretical high-energy astrophysics.

* The Restless Universe: Understanding X-ray Astronomy in the Age of
Chandra and XMM-Newton
E.M. Schlegel
Oxford University Press, 2002
ISBN: 0-19-514847-9
Fairly recent, nontechnical, broad review of X-ray astronomy.

* Black Holes, White Dwarfs, and Neutron Stars: The Physics of Compact Objects
S.L. Shapiro and S.A. Teukolsky
Wiley Interscience, 1983
ISBN: 0-471-87316-0
This somewhat old but renown book covers the physics compact objects. It is very well written in places more advanced than undergraduate course.

GW Gelman Library should have most of these books. However, if you are planning a professional career in astronomy/astrophysics, you will need to take a look at these books at some point so you may want to buy them in Amazon (used books are cheaper).

In addition to the textbooks, *I also strongly encourage you to look through the preprints abstracts http://arxiv.org/list/astro-ph/recent for articles relevant for the topic of the upcoming class. If something catches your attention, we shall discuss it at the beginning of each class period. This will count toward your active participation part the course grade.*

GRADING: Your grade will be based on homework (30%), the midterm exam (20%), class attendance/participation (20%), observing proposal term project (30%). Lectures are mandatory and active participation during the class period is expected. If you missed my lecture because of medical reasons you are required to bring a note from your doctor stating that you have been sick.

The grading scale for the course is as follows:

106.00 - 94.000	A
93.999 - 90.000	A–
89.999 - 86.000	B+
85.999 - 82.000	B
81.999 - 78.000	B–
77.999 - 74.000	C+
73.999 - 70.000	C
69.999 - 66.000	C–
65.999 - 61.000	D+
60.999 - 55.000	D
54.999 - 50.000	D–

49.999 - 0.000 F

HOMEWORK AND TEAMWORK POLICY: Homework assignments will be distributed and collected ***before*** the class starts. The due dates are specified otherwise in the course timetable (unless a different date is announced in class). The homework problems will also be available through the Homework section of the course in the Blackboard. Any homework turned in late, without a valid excuse (e.g., medical), will be assigned only half of the credit. Please start working on your homework early and do not postpone it till the last day!

In this course you are expected to think independently and creatively. You will need this skill in your future careers. You are required to work by yourself (alone!) on each problem for an hour before you discuss it with anyone else or come to the office hours for help. Feel free to consult the textbooks and lecture notes but *do not* google the solution. After you made an honest attempt to solve a problem for at least 1 hour, you may discuss it with other students. Any substantial use of online resources must be *clearly indicated* in your homework. A partial credit is given if the answer wrong or even if the wrong method is used but a reasonable hypothesis about the underlying physical processes is formulated and an effort to solve the problem is demonstrated. Should a particular problem cause troubles for majority of students in the class, it will be reviewed in class.

It is required to write the homework solution in a detailed and clear way. If the logic of the solution is unclear, no credit will be given. A detailed, well-structured solution will always result in a higher grade. Please, do not substitute the numbers into the equations until the very end, whenever possible. Clearly number the problems in your homework in the same way as they are numbered in the homework assignment. Some homework sets may include more challenging (research grade) problems for extra credit. Solving them can be used to compensate for poor performance in exams.

Each problem (homework or exams) is graded on a five-point scale. One point is awarded if you demonstrate understanding of the physical processes associated with the problem. Another point is awarded if you use the correct equations (assuming the equations are needed to solve the problem). Two points are awarded for correct solution of the equations. Final point is awarded if the numerical answer (if required) is correct or, if you got the wrong numerical answer but realized this and wrote a more reasonable answer you found elsewhere. There may deviation from this grading scheme, if the problem does not require any equation solving or numerical answers.

If you disagree with the homework grade, you must appeal by e-mail **within 1 week** from the date when graded homework was distributed. The appeal must include your name, clearly identify the issue in question, contain a detailed explanation of what you think is wrong or unfair, and your original homework must be included as a part of the appeal. The same procedure applies to the exam.

EXAMS: There will be **one (midterm) exam** and the term project (observing proposal) in lieu of the final exam. The midterm exam will cover all topics up to the day of the exam. You will be responsible for knowing the material presented in class as well as material from the assigned readings including the textbooks and other assigned reading material (unless the material was marked as optional). The exam will be closed book and closed notes (except for those distributed to you). You may use standard, non-programmable

calculators on the exams. Cell phones, laptops, Ipads, and calculators that can store equations and text are not allowed. The violation of this policy will cost you half the exam score. A table with physical constants (in cgs) will be provided during the exam.

The midterm exam will have about 5 questions/problems. Questions may include definitions, physical explanations, brief calculations, and brief derivations. You should work efficiently to earn as many points as possible. Do not get stuck on a single question for the entire exam period. An incomplete solution can result in partial credit. To get the full or partial credit, your written answers must be easily legible and must demonstrate your reasoning clearly. Please provide your solutions in the same order in which the questions are presented.

If you missed the exam and have a medical or other valid excuse, you must contact the instructor as soon as possible regarding this matter.

TERM PROJECT: Your own proposal to study a high-energy object or phenomenon with the observatory of your choice (e.g., Chandra, XMM-Newton, NuStar, Swift, Hubble, etc.) represents your independent term project. You will need to write the proposal and submit by the date specified in the course schedule (see above). You will also need to present your proposal to the rest of the class and defend it (imagine the Time Allocation Committee). Examples of successful observing proposals will be distributed after the midterm. There are some important intermediate steps in the process of proposal preparation. Please see the course schedule for details and due dates for the intermediate steps (see above).

CLASS ATTENDANCE AND PARTICIPATION: The class attendance and participation are very important. They are part of the grade (see above). To do well in this course , you should (1) come to class and pay attention during the lecture, (2) answer questions when they are posed by the instructor, (3) ask questions in class when you don't understand something, have an idea, or just curious about something related to the lecture topic, (4) perform simple tasks at the whiteboard when requested by the instructor, (5) be courteous and friendly to your fellow students and the instructor.

OFFICE HOURS AND QUESTIONS: You are *strongly* encouraged to come to the office to get help with the course. If you cannot make it at the appointed times, please e-mail the instructor to find an alternative different time to meet.

Miscellaneous

***Blackboard* system:** The *Blackboard* courseware system will be used for this course. The address for the *Blackboard* web site is: **http://blackboard.gwu.edu**

After entering *Blackboard*, it is necessary for you to click on the PHYS 6730 course. You are *automatically* subscribed within the *Blackboard* system to the courses for which you are registered (but you also must have a GW e-mail address!). The web access provided by *Blackboard* is a valuable resource for all aspects of the class. It includes course announcements, lecture notes, homework solutions, discussion forums, and other useful features. You should visit it frequently!

University policies

Use of Electronic Course Materials and Class Recordings

Students are encouraged to use electronic course materials, including recorded class sessions, for private personal use in connection with their academic program of study. Electronic course materials and recorded class sessions should not be shared or used for non-course related purposes unless express permission has been granted by the instructor. Students who impermissibly share any electronic course materials are subject to discipline under the Student Code of Conduct. Please contact the instructor if you have questions regarding what constitutes permissible or impermissible use of electronic course materials and/or recorded class sessions. Please contact Disability Support Services at disabilitysupport.gwu.edu if you have questions or need assistance in accessing electronic course materials.

Academic Integrity Code

Academic Integrity is an integral part of the educational process, and GW takes these matters very seriously. Violations of academic integrity occur when students fail to cite research sources properly, engage in unauthorized collaboration, falsify data, and in other ways outlined in the Code of Academic Integrity. Students accused of academic integrity violations should contact the Office of Academic Integrity to learn more about their rights and options in the process. Outcomes can range from failure of assignment to expulsion from the University, including a transcript notation. The Office of Academic Integrity maintains a permanent record of the violation.

More information is available from the Office of Academic Integrity at studentconduct.gwu.edu/academic-integrity. The University's "Guide of Academic Integrity in Online Learning Environments" is available at studentconduct.gwu.edu/guide-academic-integrity-online-learning-environments. Contact information: rights@gwu.edu or 202-994-6757.

University policy on observance of religious holidays

In accordance with University policy, students should notify faculty during the first week of the semester of their intention to be absent from class on their day(s) of religious observance. For details and policy, see "Religious Holidays" at provost.gwu.edu/policies-procedures-and-guidelines.

Support for students outside the classroom

Virtual academic support

A full range of academic support is offered virtually in the fall 2020 and spring 2021 semesters. See coronavirus.gwu.edu/top-faqs for updates.

Tutoring and course review sessions are offered through Academic Commons in an online format. See academiccommons.gwu.edu/tutoring

Writing and research consultations are available online. See academiccommons.gwu.edu/writing-research-help

Coaching, offered through the Office of Student Success, is available in a virtual format. See studentsuccess.gwu.edu/academic-program-support

Writing Center

GW's Writing Center cultivates confident writers in the University community by facilitating collaborative, critical, and inclusive conversations at all stages of the writing process. Working alongside peer mentors, writers develop strategies to write independently in academic and public settings. Appointments can be booked online. See gwu.mywconline.

Academic Commons

Academic Commons provides tutoring and other academic support resources to students in many courses. Students can schedule virtual one-on-one appointments or attend virtual drop-in sessions. Students may schedule an appointment, review the tutoring schedule, access other academic support resources, or obtain assistance at academiccommons.gwu.edu. Academic Commons offers several short videos addressing different virtual learning strategies for the unique circumstances of the fall 2020 and spring 2021 semesters. See academiccommons.gwu.edu/study-skills. They also offer a variety of live virtual workshops to equip students with the tools they need to succeed in a virtual environment. See tinyurl.com/gw-virtual-learning

Disability Support Services (DSS) 202-994-8250

Any student who may need an accommodation based on the potential impact of a disability should contact Disability Support Services to establish eligibility and to coordinate reasonable accommodations. disabilitysupport.gwu.edu

Counseling and Psychological Services 202-994-5300

GW's Colonial Health Center offers counseling and psychological services, supporting mental health and personal development by collaborating directly with students to overcome challenges and difficulties that may interfere with academic, emotional, and personal success. healthcenter.gwu.edu/counseling-and-psychological-services